



Jashore University of Science and Technology
Department of Computer Science and Engineering

B.Sc. Engineering Semester Final Examination-2022

2nd Year 2nd Semester, Session: 2019 - 2020

Course Code: BUS 2201

Course Title: Business Psychology

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) What do you mean by psychology? Being a student of CSE, why is it important to know psychology? 05
- b) Illustrate the nature of psychology? Clarify the relationships among business, psychology and computer science. 07
2. a) Describe the different types of endocrine glands of the human body and their effects on human behavior. 07
- b) Explain the 'Lymbic system' of the human brain and the functions of different components of 'Lymbic system'. 05
3. a) How Piaget's theory explain the cognitive development? 05
- b) Explain the levels of morality development with examples of each level. 04
- c) Discuss the 'Sternberg's Triarchic Theory' of intelligence. 03
4. a) What is perception? Discuss the nature of perception. 05
- b) Explain the steps of perception. 03
- c) Briefly explain the causes of illusion? 04
5. a) Discuss the 'Classical Conditioning Theory' of learning with the elements and key features. 07
- b) What are the aspects of 'Cognitive Learning Theory'? Explain with examples. 05
6. a) How could you apply 'McClelland's Theory' of motivation to motivate the employees of your organization? 07
- b) Explain 'Maslow's Need Hierarchy theory' of motivation? How is it applicable in a business organization? 05
7. a) Suppose you have a software development firm and are going to hire for the position 'programmer' to develop web applications, database-driven web interfaces, Design and implement user-driven templates, databases and interfaces. For the above context prepare job description, job specifications, and job evaluation. 08
- b) Explain the major techniques of job design? 04
8. a) Differentiate between recruitment and selection. 02
- b) Suppose your organization is going to hire a 'web developer', as one of the members of the selection team, suggest a selection process for the aforesaid position. 10



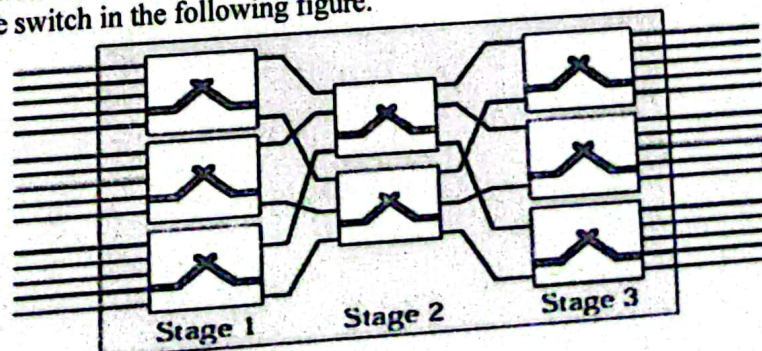
Department of Computer Science and Engineering
Jashore University of Science and Technology
Semester Final Examination-2022
2nd Year 2nd Semester, Session: 2019-20

Course Code: CSE 2205
Course Title: Data Communication

Time: 3.00 hours
Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

- 1.a) Identify the components of a Data Communication System. Write the differences between half-duplex and full-duplex transmission modes. 5
- b) Define Topology. Draw a Hybrid topology with a star backbone and four ring networks. 4
- c) Explain how signal changes its shape or form. 3
- 2.a) What is Shannon Capacity? A telephone line normally has a bandwidth of 2600 Hz. The SNR is usually 3194. Calculate the capacity of this channel. 3
- b) Define Throughput. Suppose a network with bandwidth of 20Mbps can pass only an average of 27,000 frames per minute with each frame carrying an average of 20,000 bits. What is the throughput of this network? 4
- c) What Does the Nyquist theorem have to do with communications? Suppose we need to send 290 kbps over a noiseless channel with a bandwidth of 24 kHz. How many signal levels do we need? 3
- d) What is Quadrature Amplitude Modulation (QAM)? 2
- 3.a) What is guard band? Assume that a voice channel occupies a bandwidth of 3 kHz. We need to combine six voice channels into a link with a bandwidth of 18 kHz, from 60 to 78 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands. 5
- b) What do you understand by empty slots? How to handle disparity in the data rates with TDM? 4
- c) Write the differences between radio waves and microwaves. 3
- 4.a) Discuss about various types of Propagation Methods. 3
- b) What is the significance of twisting in twisted-pair cable? 3
- c) How can we connect multiple devices in an efficient way? Explain. 3
- d) For which reason we use multistage switch? Calculate the number of cross points required by the multistage switch in the following figure. 3



- 5.a) Implement a binary ASK conversion procedure. 5
- b) Discuss about the major components of a packet switch and their functions. 4
- c) Define carrier signal and its role in analog transmission. 3
- 6.a) How can we use the Hamming Code for error correction? Explain with example. 4
- b) Suppose the following block of 16 bits is sent using a CHECKSUM of 16 bits 01101001 00010001. Show how to know either the data are accepted or rejected. 3
- c) How does a single-bit error differ from a burst error? Discuss the Two Dimensional Parity Check and types of errors it can detect. 5
- 7.a) Why is flow control needed? How are delayed acknowledgments handled in Stop and Wait ARQ? 3
- b) In a Stop and Wait ARQ system, the bandwidth of the line is 3 Mbps and 1 bit takes 30 ms to make a RTT. What is the BDP? If the system data frames are 3000 bits in length, what is the utilization percentage of the link? Compare the result with Go Back N ARQ with a 21 frame sequence. 3
- c) How lost frames are handled in Selective Repeat ARQ operations? 3
- d) Discuss how sliding window protocol works in Go-back-N ARQ operations. 3
- 8.a) Describe the multiplexing and de-multiplexing techniques of FDM. 6
- b) Distinguish between a link and a channel in multiplexing. 3
- c) Write down the characteristics of microwave propagation. 3



Jashore University of Science and Technology
Department of Computer Science and Engineering

B.Sc. Engineering Semester Final Examination-2022

2nd Year 2nd Semester, Session: 2019 - 2020

Course Code: CSE 2203

Course Title: Database Management System

Time: 3.00 hours

Marks: 72

[NB Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

- Mention the various kinds of database users. Describe sophisticated users. 03
- Define Database Instance and Database Schema. Can Database Schema ever be changed? Explain your answer. 04
- "All candidate keys are super keys, but all super keys are not candidate keys" – justify this statement 03
- Write the characteristics that distinguish the database approach with the file-based approach. 02

2. Consider the following relational schema of a Bank database – 6*2=12

Branch (branch_name, branch_city, assets)

Customer (ID, customer_name, customer_street, customer_city)

Loan (loan_number, branch_name, amount)

Borrower (ID, loan_number)

Account (account_number, branch_name, balance)

Depositor (ID, account_number)

Assume that branch names and customer names uniquely identify branches and customers, but loans and accounts can be associated with more than one customer.

You are required to (use relational algebra for III, IV, V and VI):

- Find the appropriate primary keys.
- Given your choice of primary keys, identify appropriate foreign keys.
- Find the name of each branch located in "Jashore".
- Find each loan number with a loan amount greater than 10000/-.
- Find the ID of each depositor who has an account with a balance greater than 6000/-.
- Find the ID of each depositor who has an account with a balance greater than 6000/- at the "JUST" branch.

3. Consider the following relational schema of an employee database – 6*2=12

Employee (employee_name, street, city)

Works (employee_name, company_name, salary)

Company (company_name, city)

Manages (employee_name, manager_name)

You are required to write Structured Query Language to:

- Find the names of all employees who work for First Bank Corporation.
- Find all employees in the database who live in the same cities as the companies for which they work.
- Find all employees in the database who live in the same cities and on the same streets as do their managers.

IV. Find all employees who earn more than the average salary of all employees of their company.

V. Find the company that has the smallest payroll.

iii) Give all managers of First Bank Corporation a 10 percent raise in salary.

4. a) Note that an employee may simply have no manager listed or may have a null manager in the employee database in question no 3. 3+3+3

I. Write a query to find those employees with no manager.

II. Write your query using an outer join and then write it again using no outer join at all.

b) Consider the same schema in question no 3. 03

Using SQL create a view *tot_salary* (*company_name*, *total_salary*), giving the total salary employees of each company get.

5. a) Define functional dependency. Describe different types of functional dependency with proper example. 04

b) Define indexing. Differentiate between sparse indexing and dense indexing. 03

c) Consider the following schema 05

Student (*RollNo*, *FirstName*, *EnglishMarks*, *PhysicsMarks*, *ChemistryMarks*, *MathsMarks*, *TotalMarks*);

You are required to:

Create a Trigger for each row to calculate and set the value of *TotalMarks* while inserting value of other columns. Please note that $TotalMarks = EnglishMarks + PhysicsMarks + ChemistryMarks + MathsMarks$

6. Design a database for an automobile company to provide to its dealers to assist them in maintaining customer records and dealer inventory and to assist sales staff in ordering cars. Please note that: 4+4+4

- Each vehicle is identified by a vehicle identification number (VIN).
- Each individual vehicle is a particular model of a particular brand offered by the company (e.g., the XF is a model of the car brand Jaguar of Tata Motors).
- Each model can be offered with a variety of options, but an individual car may have only some (or none) of the available options.
- The database needs to store information about models, brands, and options, as well as information about individual dealers, customers, and cars.

Your design should include:

I. An E-R diagram,

II. A set of relational schemas (derived from E-R diagram)

iii) A list of constraints, including primary-key and foreign-key constraints

7. a) Normalize the following schema, with given constraints, to 4NF. 06

books (*accession_no*, *isbn*, *title*, *author*, *publisher*)

users (*user_id*, *name*, *dept_id*, *dept_name*)

$accession_no \rightarrow isbn$

$isbn \rightarrow title$

$isbn \rightarrow publisher$

$isbn \twoheadrightarrow author$
 $user_id \twoheadrightarrow name$
 $user_id \twoheadrightarrow dept_id$
 $dept_id \twoheadrightarrow dept_name$

b) Consider the following set F of functional dependencies on the relation schema r . 3*2=06

$r(A, B, C, D, E, F)$

$A \rightarrow BCD$

$BC \rightarrow DE$

$B \rightarrow D$

$D \rightarrow A$

You are required to:

I. Compute a canonical cover for the above set of functional dependencies F ; give each step of your derivation with an explanation.

II. Compute a 3NF decomposition of r based on the canonical cover.

III. Compute a BCNF decomposition of r using the original set of functional dependencies.

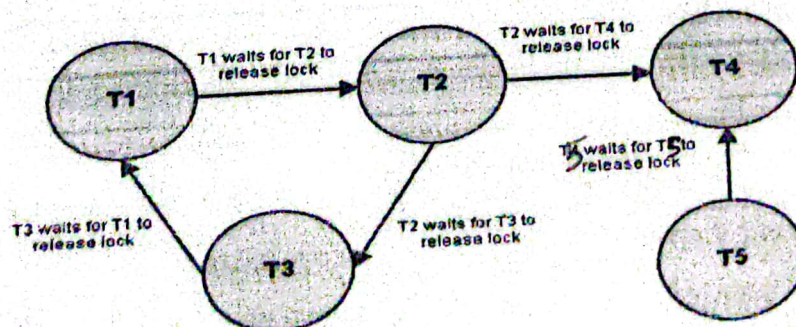
8/ a) Identify, which of the following situations of transactions T3 and T4, database consistency will be preserved in? 03

I. Both of T3 and T4 want to read table X.

II. T3 wants to read but T4 wants to write the table Y.

III. T4 wants to read a table named Z while T3 tries to Write another table P

b) Explain the following wait-for-lock graph and determine whether any deadlock prevails here or not. T1, T2, T3, T4 and T5 are transactions. 04



c) Determine whether the following schedule of transactions T1 and T2 is conflict serializable or not. 05

T1	T2
Read (B)	Read (B)
$B \leftarrow B + 1$	$B \leftarrow B + 1$
Write (B)	Write (B)
Commit	Commit



Jashore University of Science and Technology
Department of Computer Science and Engineering

B.Sc. Engineering Semester Final Examination-2022

2nd Year 2nd Semester, Session: 2019 - 2020

Course Code: MATH 2201

Course Title: Probability and Statistics for Engineers

Time: 3.00 hours

Marks: 72

[NB. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks]

1. a) Define Harmonic Mean. For a group of 200 candidates, the mean and standard deviation of scores were found to be 40 and 15 respectively. Later on, it was discovered that the scores 43 and 35 were misread as 34 and 53 respectively. Find the corrected mean and standard deviation corresponding to corrected figures. 1+3

- b) Define Moments. Calculate: (i) Quartile Deviation (Q.D), and (ii) Mean Deviation (M.D) from mean, for the following data: 1+3

Marks.	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70
No. of Students	6	5	8	15	7	6	3

- c) Calculate the first four moments of the following distribution about the mean and hence find β_1 and β_2 . 04

x:	0	1	2	3	4	5	6	7	8
f:	1	8	28	56	70	56	28	8	1

2. a) Define Statistics. Write the limitations of statistics and describe these. 1+3

- b) The median and mode of the following wage distribution are known to be Rs. 3,350 and Rs. 3,400 respectively. Find the values of f_3 , f_4 and f_5 . 04

Wages (in Rs.)	No. of Employees	Wages (in Rs.)	No. of Employees
0 - 1,000	4	4,000 - 5,000	f_5
1,000 - 2,000	16	5,000 - 6,000	6
2,000 - 3,000	f_3	6,000 - 7,000	4
3,000 - 4,000	f_4	Total	230

- c) Define Arithmetic Mean. The average salary of male employees in a firm was 5,200 BDT and that of females was 4,200 BDT. The mean salary of all the employees was 5,000 BDT. Find the percentage of male and female employees. 1+3

3. a) Define Bayes' theorem. Describe the theorem with proper equation. 1+3

- b) The probabilities of X, Y and Z becoming managers are $\frac{4}{9}$, $\frac{2}{9}$ and $\frac{1}{3}$ respectively. The probabilities that the Bonus Scheme will be introduced if X, Y and Z becomes Managers are $\frac{3}{10}$, $\frac{1}{2}$ and $\frac{4}{5}$ respectively. 04

(i) What is the probability that Bonus Scheme will be introduced, and

(ii) if the Bonus Scheme has been introduced, what is the probability that the manager appointed was X?

- c) A, B and C are three mutually exclusive and exhaustive events associated with a random experiment. Find $P(A)$ given that $P(B) = \frac{3}{2} P(A)$ and $P(C) = \frac{1}{2} P(B)$. 02

- d) Two dice are tossed. Find the probability of getting an even number on first die or a total of 8. 02

- ✓ a) Write the merits and demerits of Mode. Write the properties of Arithmetic Mean. 2+2
- b) In a factory employing 3,000 persons, in a day 5 percent work less than 3 hours, 580 work from 3.01 to 4.50 hours, 30 percent work from 4.51 to 6.00 hours, 500 work from 6.01 to 7.50 hours, 20 percent work from 7.51 to 9.00 hours and the work 9.01 or more hours. What is the median hour of work? 04
- c) Define Geometric Mean. Prove that, $\log G = (n_1 \log G_1 + n_2 \log G_2) / (n_1 + n_2)$. Where, n_1 and n_2 are the sizes, G_1 and G_2 the geometric means of two series, the geometric mean G . 1+3

5. a) Define normal distribution. Find its mean and variance. 04
- b) If X is distributed as binomial, show that $\text{mean} \geq \text{variance}$. 04
- c) Let three unbiased coins are tossed at a time. What is the probability that (i) no head, (ii) at least two heads will appear? 04

- ✓ 6. a) If X and Y are independent random variables, then prove that, $E[h(X), k(Y)] = E[h(X)] E[k(Y)]$, where $h(\cdot)$ and $k(\cdot)$ is the function of X and Y alone respectively. 03
- b) Calculate the correlation coefficient for the following heights (inches) of fathers (X) and their sons (Y): 03

X:	65	66	67	67	68	69	70	72
Y:	67	68	65	68	72	72	69	71

- c) Define Probable Error (P.E). Write the properties of regression Coefficients. 03
- d) Find the most likely price in Dhaka corresponding to the price of BDT 70 at Khulna from the following: 03

	Khulna	Dhaka
Average price	65	67
Standard deviation	2.5	3.5

Correlation coefficient between the prices of commodities in the two cities is 0.8

- ✓ 7. a) The probability of a man hitting a target is $1/4$: 04
- (i) If he fires 7 times what is the probability of his hitting the target at least twice?
- (ii) How many times must he fire so that the probability of his hitting the target at least once is greater than $2/3$?
- b) Define Poisson Distribution. In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error. 04

- c) Write the Characteristics of normal distribution and normal probability curve. 04

8. a) Define correlation coefficient, partial correlation coefficient, and multiple correlation coefficient. 03

- b) How do you interpret the range of correlation coefficient? 03

- c) Find the value of correlation coefficient from the following table: 03

Age(X)	43	21	25	42	57	59
Sugar Level(Y)	99	65	79	75	87	31

- d) If X and Y are independent show that they are uncorrelated. 03

Department of Computer Science and Engineering
Jashore University of Science and Technology
2nd Year, 2nd Semester, Final Exam: 2022

Course: CSE-2207

Time: 3.00 hours

Course Title: Software Engineering

Marks: 72

[N.B.: Answer any Six (06) sets of questions from the following eight (08) sets of questions. The numbers at the right side of each question indicate assigned marks.]

01. a) Write the definition of software engineering. What are the key challenges facing in software engineering? 03
- b) Explain the differences between software engineering and computer science. 02
- c) Briefly discuss the four fundamental activities of software engineering. 04
- d) What are the two types of software products? Present at least one example for each type. 03
02. a) Describe the software process framework with its activities like a tree-based structure. 04
- b) All the process models have the same to develop a software in a better way. Then why there are so many software process models and what are their significance? 04
- c) What is the difference between plan-driven and agile process model and describe the process flow of scrum model 04
03. Consider an online flight booking system with the following requirements-
- Every online booking needs to be associated with an account
 - One account cannot be associated with multiple users
 - Use of captcha and encryption to avoid bots from booking tickets
 - Search results should enable users to find the most recent and relevant booking options
 - Search results should populate within acceptable time limits
 - System should enable users to book / pay for their tickets only in a timeboxed manner after tickets being added to the cart
 - User should be helped appropriately to fill in the mandatory fields, in case of invalid input
 - System should only allow users to move to payment only when mandatory fields such as date, time, location has been mentioned
 - System should consider time zone synchronization when accepting bookings from different time zones
 - Booking confirmation should be sent to user to the specified contact details
 - System should visually confirm as well as send booking confirmation to the user's contact
 - System should accept payments via different payment methods, like PayPal, wallets, cards, vouchers, etc.
- a) Identify the functional and non-functional requirements. 03

- b) Design a set of use case diagram based on the above scenario. 05
- c) Write the user story for one of the main components of the above-mentioned system. 04✓
- ~~04.~~ a) Discuss some of the problems that occur when requirements must be elicited from three or four different customers. 03
- b) Write the questions that are asked in the time of requirement validation. 04
- c) Draw the class diagram considering the "Question 3" requirements. 05
- ~~05.~~ a) Is it possible to begin coding immediately after a requirements model has been created? Explain your answer, and then argue the counterpoint. 04✓
- b) Explain the importance of Unified Model Language (UML) in the context of software engineering life cycle. 03
- ~~06.~~ c) Develop the activity diagrams for the system mentioned in "Question 3". 05✓
- a) What are the 4Ps in software project management. 02✓
- b) Describe the W5HH principles for software project and processes. 04✓
- ~~07.~~ c) Team A found 342 errors during the software engineering process prior to release. Team B found 184 errors. What additional measures would have to be made for projects A and B to determine which of the teams eliminated errors more efficiently? What metrics would you propose to help in making the determination? What historical data might be useful? 06
- a) You have been appointed a project manager for a small software products company. Your job is to build a breakthrough product that combines virtual reality hardware with state-of-the-art software. Because competition for the home entertainment market is intense, there is huge pressure to get the job the done. What team structure would you choose and why? What software process model(s) would you choose and why? 04
- b) What are the attributes of effective software metrics? Briefly describe these attributes. 02
- c) A system has 32 user inputs, 60 external outputs, 24 fields for different user queries, manages 8 internal logical files, and interfaces with 2 different legacy systems. Compute the function point value for the system. Assume that, all complexity adjustment values are average. 03✓
- ~~08.~~ a) Describe different types of risk in software engineering. 04✓
- b) Draw the RMMM plan the system mentioned in "Question 3". 05✓
- c) Draw the project scheduling for the system mentioned in "Question 3".



Department of Computer Science and Engineering
Jashore University of Science and Technology
Semester Final Examination-2022
2nd Year 2nd Semester, Session: 2019-20

Course Code: CSE 2201

Course Title: Software Design Pattern

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

1. (a) Discuss encapsulation and Polymorphism with examples 4
(b) Design a program to handle exceptions for student id. 5
i.e. student id is composed of someone's admission_year-dept_id-serial_no (2020-01-01) so all the id requests without the correct format will be caught by an 'unformatted id' exception.
(c) What is the difference between an abstract class and an interface 3
2. Consider a property owner sells three types of houses
- General houses: Composed of a boundary wall, gate, building materials, number of rooms
- Apartment building: Composed of a boundary wall, gate, number of rooms in each apartment, number of apartments, stairs
- Bungalow: Composed of a boundary wall, gate, number of rooms in each apartment, number of apartments, stairs, swimming pool, and tennis ground.
You are assigned to design a software-based solution for the property business as a software engineer.
(a) Design the software solution for the described problem using a design pattern. Clearly draw the classes and describe their responsibilities 6
(b) Which design pattern suits best for the problem? Why 3
(c) How can you utilize a "Director" in the above problem? Explain the necessity of "Director" in such a solution. 3
3. (a) State the name of a design pattern that restricts a class to have only one instance. Write down the ideas how this pattern restricts a class from creating multiple instances. 4
(b) Write the major differences between shallow copy and deep copy. Also mention the name of a design pattern that provides a way of object cloning. 4
(c) List down the participants of abstract factory pattern. Briefly explain the role of each participant. 4

4. Consider a university admission system that consists of migrations. After each of migration, a set of students are assigned to a department. You are assigned to manage the migration system in such a way that, migrations are versioned in a serialized way and you can restore any version of the migration to restore and check for internal audits.
- (a) Identify a design pattern to design the solution for the above system. Explain the solution with the help of classes and other infrastructures. 6
- (b) Implement the solution with object-oriented computer programming. Demonstrate at least two separate audits. 5+1
5. (a) What is the "Factory" design pattern? Which scenario best suits to implement the "Factory" design pattern? Explain with an example. 2+3
- (b) Discuss the differences between "Factory" and "Abstract Factory" design patterns. 4
- (c) Justify the claim that "Abstract factory is a Factory of the Factories" 3
6. (a) Describe the proxy design pattern with an example problem and corresponding solution. 7
- (b) What are the differences between the Adapter and the Bridge design pattern? 5
7. Match each pattern with its description with proper Justification: 12

Patterns	Scenarios
• Adapter	a. Wraps another object and provides a different interface to it
• Iterator	b. Wraps another object to control access to it
• Facade	c. Wraps a bunch of objects to simplify their interface
• Observer	d. Subclasses decide which concrete classes to create
• Proxy	e. Ensures one and only one object is created
• Factory	f. Provides a way to traverse a collection of objects without exposing its implementation
• Singleton	g. Allows objects to be notified when state changes
• Builder	h. Decouples an abstraction from its implementation so they can vary independently
• Mediator	i. Provides the ability to restore an object to its previous state
• Memento	j. Encapsulates how a set of objects interact
• Bridge	k. Allows constructing in steps
	l. Allows a client to create families of objects without specifying their concrete classes

N.B: One pattern may match with different scenarios

8. (a) What are the SOLID principles? Explain each of the items with an example. 8
- (b) Identify the differences between design patterns and design principles. 4